

Day 1 - A/B Testing

A/B Testing Process -

- Define Hypotheses
- Check assumptions
- Calculate CTR
- Statistical test
- Confidence interval
- Statistical power

- A/B testing falls under Inferential statistics.

Define Hypothesis:

- Control Group: - The current or default version of a product, feature, or website.
 - It serves as a baseline to measure what "normal" performance looks like. (no change)
- Treatment Group: - The group exposed to the new variation (feature, design) being tested.
 - Shows how the change affects user behavior.
- Null Hypothesis: - The statement of "No Change"
 - The CTR for "AI Engineer" is equal to or less than the CTR for "Data Scientist" (no improvement)
- Alternate Hypothesis: - The statement of "Change"
 - The CTR for "AI Engineer" is greater than the CTR for "Data Scientist" (positive improvement)
- One sided test: - If the concern is more about difference ^{direction} "tve" or "-ve"
- Two sided test: - If the concern is more about difference "tve" or "-ve"

Check Assumptions :

- Independence : Each user's decision to click is independent of the others.
- Sample size : The sample should be large enough to invoke the central Limit Theorem (typically > 30 per group).
- The CLT states that the distribution of the sample mean approaches a normal distribution as the sample size grows, regardless of the population's original distribution. This is fundamental in hypothesis testing & confidence intervals.
- Distribution : The distribution of clicks follows a binomial distribution, which, for large samples, approximates a normal distribution.
- A binomial distribution models the no. of successes (e.g. clicks) in a fixed no. of independent trials (e.g., ~~clicks~~ impressions), where each trial has only two possible outcomes (success or failure).
- Binomial falls into discrete probability distribution.

Calculate CTR : (Click through rate).

- To check the performance of each Job Group :
 - CTR for Data Scientist : $\text{clicks} / \text{Impressions}$
 - CTR for AI Engineer : $\text{clicks} / \text{Impressions}$.

Statistical Test :

- We use a Z-test for proportions to determine if the difference in CTR b/w the two groups is statistically significant.
- Due to chance : If the null hypothesis is true indirectly it is due to chance.

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Statistically significant : When we are rejecting the null hypothesis.

Types of Statistical Test :

- Z-Test for proportions
- One Sample T-test
- Two Sample T-test
- Paired T-test
- Z-Test for Means
- ANOVA
- Chi-square test
- Fishers Exact test
- Mann-Whitney U Test

- If $p < 0.05$ - Reject H_0 (Null Hypothesis) | $p > 0.05$ - Accept Null Hypothesis
- "P" value states whether the things by chance, is it happened randomly or it happened really.
- Z-Stats states how far it is the standard deviation from the mean.

- Z-Statistic tells us how many standard deviations the observed difference is from the null hypothesis.
- P-value shows the probability of observing such a difference by chance if the null hypothesis is true.

Statistical Power :

- Statistical power is the probability of correctly rejecting the null hypothesis when it's false. Low power increases the risk of a Type II error (failing to detect a true difference).
- Type I Error (False Positive): Rejecting the null hypothesis when it is true.
- Type II Error (False Negative): Failing to reject the null hypothesis when it is false.